



CUSTOMS

Operator manual. An agentic launch command center that clears your commercial for fifteen markets, cites the statute behind every finding, and re-renders what fails.

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One spot, dozens of markets, and the public as your QA team

Global brands ship a single commercial into dozens of markets and check maybe six of them. The defence today is a cultural consultant, a focus group and luck.

The regulated half is concrete and already carries an invoice. France bans alcohol advertising on television under Loi Evin, and the Toubon law constrains on-screen language. Quebec bans advertising directed at children under 13 outright. The UK requires pre-clearance of every TV ad through Clearcast. Nigeria requires ARCON pre-vetting. Direct-to-consumer pharma advertising is lawful in two countries on earth.

The unregulated half costs more. Pepsi and Kendall Jenner. Dolce & Gabbana in China. H&M's "coolest monkey" hoodie. Burger King Vietnam eating with chopsticks. Every one of those was caught by the public rather than by a process, at a cost measured in tens of millions.

	TODAY	WITH CUSTOMS
Coverage	A handful of markets, manually	15 markets in one run, extensible by YAML
Evidence	A consultant's opinion	A statute citation on every finding, checked live
Timeline	Days to weeks	Minutes
Monitoring	A PDF report	A Grafana panel the crew builds and updates itself
The fix	Re-edit by an agency	Re-letter, re-prop, re-voice; verified by re-observation
The hard cases	Silently "fixed"	Refused, cited, routed to a human

Pilots get a pre-flight checklist and an instrument panel. Ad launches get luck. Nobody has an instrument for this. Customs is the instrument.

A decision, and the version that can ship

You load a commercial and a target market list, and you watch it clear customs in real time. The output is two things: a decision (can this air here, and on what evidence) and a localized master (the version that can).

15

market packs spanning distinct regulatory and cultural regimes, from Loi Evin to CONAR to RTUK

114

rules, every one naming a real statute, broadcaster code or documented sensitivity in its basis

18

observation dimensions in the fixed taxonomy, from alcohol to photosensitivity to gestures

5/6

planted landmines caught at the milestone gate, zero empty rationales, zero false positives on control shots

339

offline tests green; a live suite runs the same checks against real Gemini and a real Grafana stack

3

directions of Grafana traffic: the crew writes in, Grafana triggers the crew, the crew reads back

The demo asset failed by its own maker

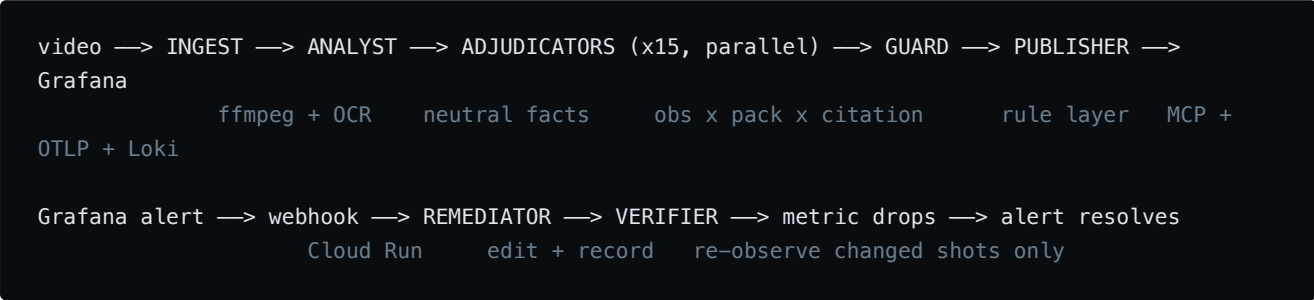
The test commercial was generated with Veo and deliberately loaded with eight documented landmines: a wine toast, an English-only handwritten tagline, beach swimwear, a strobe transition, a comparative claim, a same-sex couple holding hands, and two clean control shots. The ground truth lives in `docs/samples/landmines.yaml`. Google's own tools made the ad, and then failed it.

What it deliberately does not do

When a rule's basis targets a protected characteristic (sexual orientation, gender identity, religion, race, ethnicity, disability, caste), the Guard takes auto-remediation off the table, states the reason, and renders the finding as a human decision. Customs will tell you that you cannot ship into a market without censoring who appears in your ad, and it will not do that for you.

Observe once, judge per market

The architectural decision everything hangs off: a multimodal Analyst watches the film one time and emits neutral, timecoded observations. Per-market Adjudicators judge that single fact set against their own rulebook, in parallel.



A finding is always a join: **observation x market rule x citation**. The expensive multimodal pass happens once, not once per market; adjudicators are cheap, parallel, text-only agents; and when a brand disputes a finding, "is this fact wrong" and "is this rule wrong" are separable questions.

AGENT	INPUT	OUTPUT
Ingest	video file	shot table, keyframes, audio, transcript, OCR text
Analyst	shot, keyframes, audio	neutral observations across 18 dimensions, no verdicts
Adjudicator	observations + one market pack	findings with class, severity, rationale, grounded citation
Guard	findings	classified findings; protected-basis remediation blocked
Publisher	findings	Grafana metrics, logs, dashboards, annotations, alert rules
Remediator	one alerting finding	edited media segment plus a change record
Verifier	localized master	the finding cleared, or reopened, with evidence

The Verifier closing back onto the Analyst is what makes this a crew rather than a pipeline: remediation is not trusted until the same instrument that found the problem agrees it is gone. Unsourced citations cap a finding at severity 40 and never trigger remediation or block a market.

Vendor rule, enforced

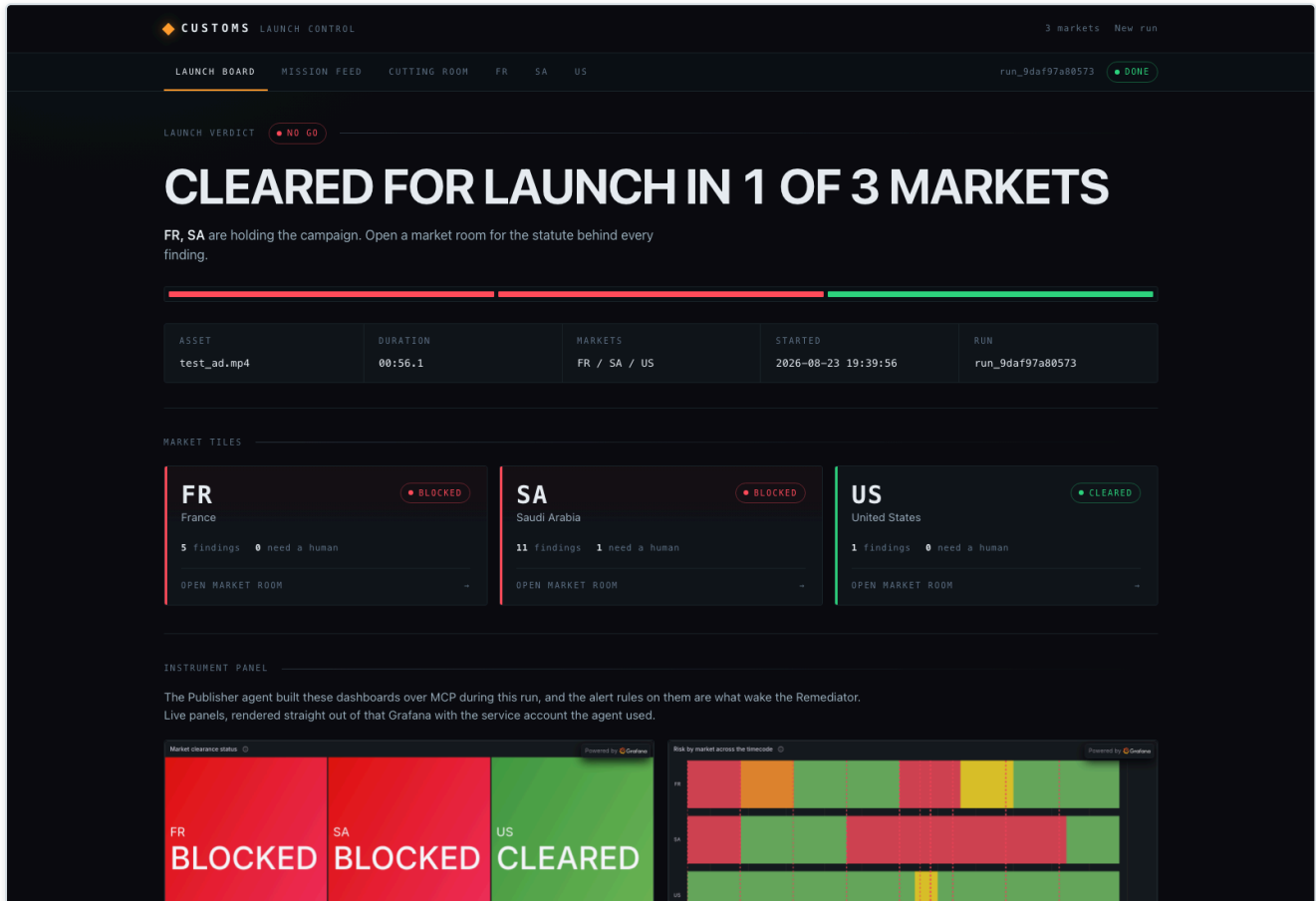
No Anthropic, OpenAI, AWS or Microsoft model, agent framework or AI API appears at runtime or in the dependency tree; the contest rule is enforced by `tests/test_no_forbidden_vendors.py`. Models are Gemini 3.x (vision, text, grounding), Gemini image editing, Gemini TTS and Veo, all via Vertex AI. Agents are coded agents on Google ADK.

Every module and where it lives

COMPONENT	ROLE	PATH
Ingest	ffmpeg shot detection, keyframes, audio; Gemini transcription and OCR	src/customs/media.py
Analyst	one multimodal pass per shot, neutral observations, no verdicts	src/customs/analyst.py
Adjudicators	observations x pack, Google Search grounding for citations	src/customs/adjudicate.py
Guard	pure rule layer; protected-basis and offence-class blocking	src/customs/guard.py
Telemetry	OTLP metrics to Mimir, log push to Loki, timecode mapping	src/customs/telemetry.py
Grafana ops	MCP client with HTTP fallback; dashboards, annotations, alerts, embeds	src/customs/grafana_ops.py
ADK crew	the agent graph; the one module importing google.adk	src/customs/crew.py
Remediator	re-lettering, prop substitution, re-voicing; change records	src/customs/remediate.py
Verifier	re-runs the instrument on changed shots; resolves or reopens	src/customs/verify.py
Console	FastAPI + Jinja2 + SSE; four screens; alert webhook	src/customs/app.py
Run store	SQLite run store and mission event bus	src/customs/store.py
Market packs	15 YAML packs, 114 rules, plus the dimension taxonomy	markets/
Dashboards	the six dashboard definitions the Publisher provisions	grafana/dashboards/
Pipeline CLI	end-to-end clearance from the terminal	scripts/run_pipeline.py
Test ad generator	Vevo commercial with documented landmines	scripts/make_test_ad.py
Deployment	Cloud Run, single instance, Secret Manager	Dockerfile , scripts/deploy.sh

Launch Board

The front door. Upload an ad, pick markets, and the board is the campaign's go or no-go.

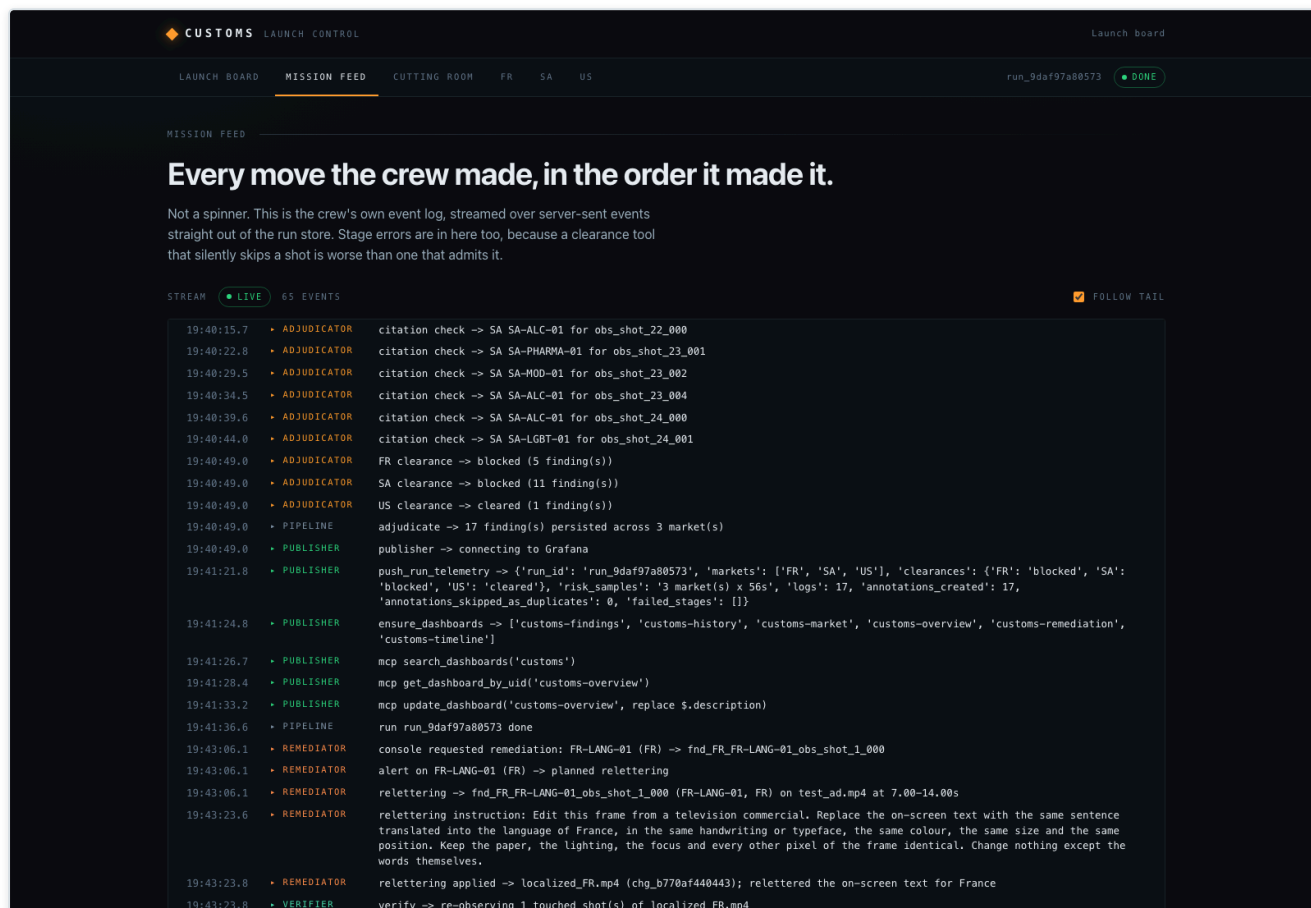


A finished FR / SA / US run: one cleared, two blocked, verdict NO GO.

- ◆ **Verdict and headline.** "Cleared for launch in 1 of 3 markets" is derived from findings, never modelled: any unresolved legal finding at severity 70 or above blocks a market; policy findings at 70+ mark it at risk.
- ◆ **Market tiles.** Pending, cleared, at risk or blocked, flipping live as each adjudicator returns. Each tile counts findings and how many need a human.
- ◆ **Instrument panel.** The embedded Grafana panels underneath (market clearance status, risk-by-timecode heatmap) are the ones the Publisher built during this very run.

Mission Feed

Every move the crew made, in the order it made it, streamed over server-sent events straight out of the run store. Not a spinner.



Adjudicator citation checks, clearance verdicts, MCP tool calls, the remediation instruction, the Verifier's re-observation.

- ◆ **Real tool calls.** `mcp update_dashboard('customs-overview')` and `push_run_telemetry` are the actual operations, not decoration.
- ◆ **Stage errors surface here too.** A clearance tool that silently skips a shot is worse than one that admits it; failed stages are published as `customs_stage_error` and rendered, never swallowed.

Market Room

One market in full: regulator, pre-clearance regime, every finding with its statute and a live citation link, and the Guard's blocked items presented as human decisions.

SA Saudi Arabia

11 FINDINGS PRE-CLEARANCE: ADVISORY GAM / SFDA

GUARD Human decision required

The guard took auto-remediation off the table for 1 finding in this market. A rule written on a protected basis, or an offence-class rule, is not something this system edits its way around: it names the problem, cites it, and hands it to a person.

SA-LGBT-01 90 00:42.1 to 00:49.1

LEGAL SOURCED

rule basis targets a protected characteristic; human decision required

Affectionate hand-holding between two men depicts a same-sex relationship, violating GAM Media Content Standards regarding Islamic values and public morals.

GAM Media Content Standards: media and advertising content contradicting Islamic values and public morals is prohibited, applied by the regulator to content perceived as condoning or promoting same-sex relationships or LGBTQ identity

REMEDiate SEND TO LEGAL REVIEW ACCEPT THE RISK

FINDINGS

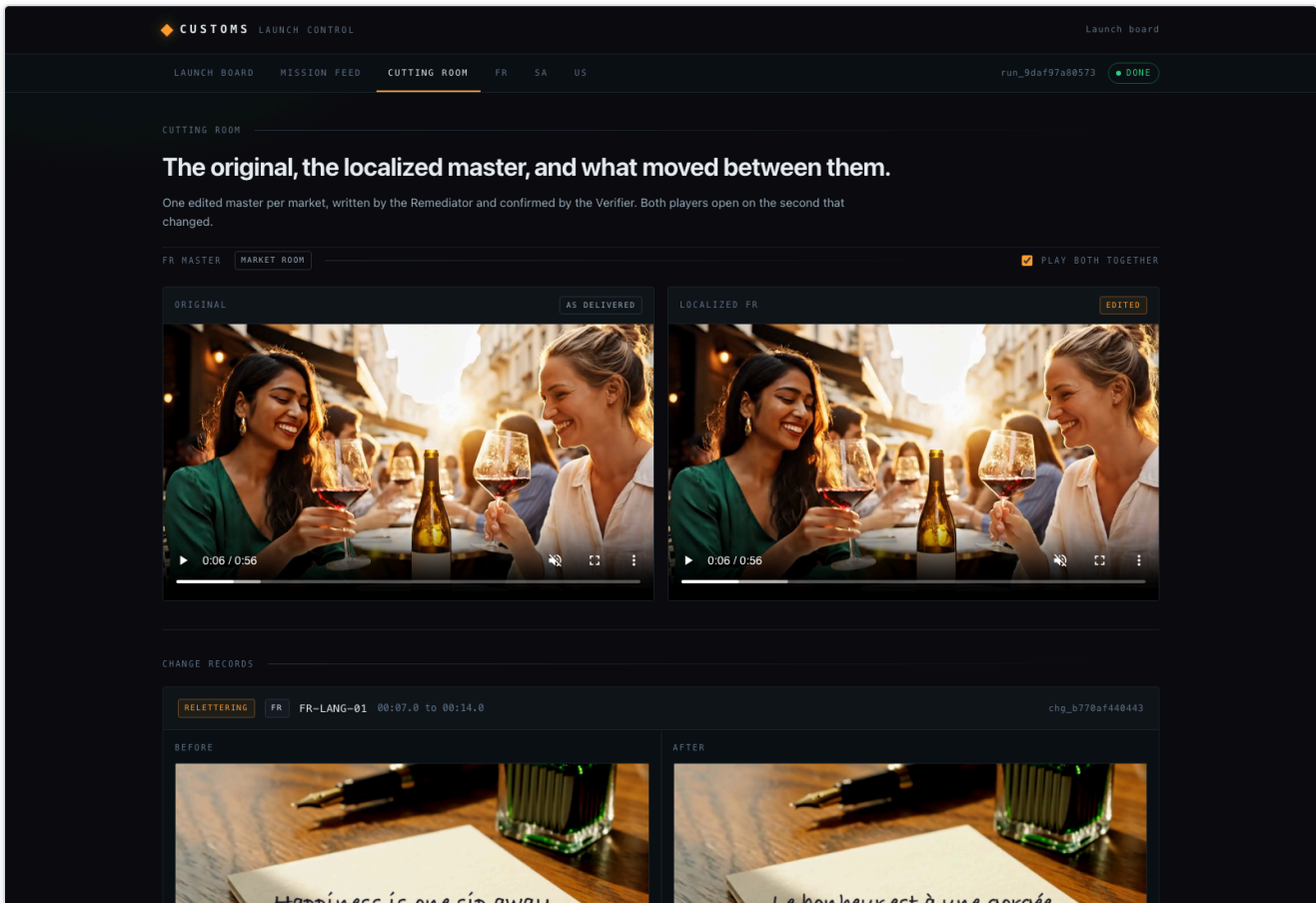
RULE	CLASS	SEVERITY	WINDOW	SOURCE	STATUS
SA-ALC-01 The explicit depiction and consumption of red wine and a wine bottle directly violates GAM Media Content Standards prohibiting the depiction or promotion of alcohol. <u>General Authority for Media Regulation (GAM, formerly the General Commission for Audiovisual Media, GCAM) Media Content Standards: advertising content may not depict or promote alcohol</u>	LEGAL	95	00:00.0 to 00:07.0	SOURCED	OPEN
SA-ALC-01 Holding a bottle associated with beer presents alcohol-related imagery,	LEGAL	95	00:21.0 to 00:28.0	SOURCED	OPEN

Saudi Arabia: 11 findings, and the Guard holding SA-LGBT-01 for a human, with the regulation cited.

- ◆ **Human decision required.** SA-LGBT-01 fires on a same-sex couple holding hands. The rule's basis targets a protected characteristic, so the Guard blocks auto-remediation and offers three honest actions: remediate anyway (a human choice), send to legal review, accept the risk.
- ◆ **Every finding cites.** Class, severity, timecode window, sourced flag, and the underlined citation resolved through Google Search grounding at adjudication time.

Cutting Room

The original, the localized master, and what moved between them. One edited master per market, written by the Remediator and confirmed by the Verifier.



FR master beside the original; both players open on the second that changed, with the change record underneath.

- ◆ **Nothing is edited silently.** Every edit produces a change record with the source finding, the method, the frames touched, and before and after stills.
- ◆ **Play both together** locks the two players so an operator can eyeball that only the offending element moved.

Alert, edit, verify, resolve

The French pack flags the handwritten English tagline under the Toubon law (FR-LANG-01). What happens next is the whole product in one loop.



BEFORE / "Happiness is one sip away", English only.



AFTER / re-lettered in French, same paper, same light.

1 The alert fires.

`customs_blocking_finding` trips on the FR series; Grafana webhooks the Cloud Run service. The webhook looks the finding up in the run store; it never trusts the payload body.

2 The Remediator plans and edits.

It reads campaign history over MCP first, then instructs Gemini image editing to translate the text in context: same handwriting, same colour, same position, change nothing except the words.

3 ffmpeg recomposites.

The edited frames are tracked across the shot and rendered into a localized FR master; a change record is written.

4 The Verifier re-observes.

The same Analyst and Adjudicator run again on the changed shots only. If the rule no longer fires, the finding resolves; anything new the edit surfaced is recorded and guarded like any other finding.

5 Grafana closes its own alert.

The metric drops, the alert resolves, and the resolving annotation lands on the run's timeline. The loop is closed by evidence.

Grafana is a participant, not a picture

Traffic runs in three directions, which is what makes the integration structural rather than decorative.

- 1

The crew writes into Grafana.

The Publisher provisions six dashboards and the alert rules over the Grafana MCP server, pushes `customs_risk` per video-second to Mimir over OTLP, finding detail to Loki with labels `{asset, market, class, dimension, rule_id}`, and writes every finding as an annotation.
- 2

Grafana triggers the crew.

Alert rules fire a webhook contact point aimed at the Cloud Run service, which wakes the Remediator. Grafana sits upstream of the work, not in a report after it.
- 3

The crew reads Grafana back.

Before remediating, the Remediator queries campaign history over MCP: has this brand tripped this rule before, and what was done about it.

DASHBOARD	WHAT IT SHOWS
Clearance Overview	per-market status tiles, blocking counts, worst offending dimension
Timeline	market by timecode heatmap; the centrepiece panel
Findings	rule, class, severity, citation link, sourced flag, from Loki
Market Detail	one market: regulator, rules, findings with statutes
Remediation	changes, before and after frames, blocked items and why
Campaign History	across assets and time; has this brand tripped this rule before

The timecode trick

Prometheus rejects backdated samples, so each run maps its timecode onto the wall clock at run start: video second `n` is written at `t0 + n`, and every panel is pinned to the run's window. The x-axis reads as the timecode because it is the timecode, offset by a constant. The run's `t0` rides on the run record and on every Loki line so panels and drill-downs agree.

Embeds use Grafana public dashboards (tokenised `d-sole` URLs) with a server-side image-render fallback behind one interface, because an iframe cannot carry a bearer token.

A market is a YAML file, not code

Fifteen packs with real citations beat a hundred and ninety-five without. The pack format is the extension point.

```
market: FR
name: France
regulators: [ARPP, Arcom]
pre_clearance: advisory      # none | advisory | mandatory
rules:
  - id: FR-ALC-01
    dimension: alcohol_tobacco_drugs
    class: legal              # legal | policy | offence
    severity: 95
    trigger: "Depiction of alcoholic drink in an advertisement"
    basis: "Loi Evin, Code de la sante publique art. L3323-2"
    source_hint: "legifrance.gouv.fr L3323-2"
    remediable: true
```

Adding market sixteen

1 Copy a pack.

Start from the closest regulatory regime in `markets/`.

2 Write real rules.

Every rule needs a globally unique `id`, a `dimension` from `markets/_taxonomy.yaml`, a class, a severity, and a `basis` naming the actual statute or code. Verify each basis with a grounded search while writing.

3 Mark protected bases honestly.

Set `protected_basis: true` where a rule targets who someone is. The Guard reads only that flag, never model output.

4 Drop it in.

The loader validates on startup; the market appears in the console on the next run. The pack test suite enforces unique ids, minimum rule counts, and that at least 40 percent of all rules are legal-class with a statute.

The launch pack spans distinct regimes rather than the largest economies: US, UK, France, Germany, Quebec, Brazil, Saudi Arabia, UAE, Turkey, India, China, Indonesia, Japan, Thailand, Nigeria.

Cold clone to running instance

Prerequisites: Python 3.12; ffmpeg on PATH; a Google Cloud project with Vertex AI enabled and `gcloud auth application-default login` done; a Grafana Cloud stack (free tier) with a service-account token (Admin) and an access-policy token (`metrics:write` , `logs:write`); the mcp-grafana v1.1.0 release binary at `bin/mcp-grafana` .

```
git clone https://github.com/tdries/customs.git && cd customs

# 1. environment
python3.12 -m venv .venv
.venv/bin/pip install -r requirements.txt
cp .env.example .env          # fill in Grafana + Google Cloud values

# 2. provision the Grafana surface (idempotent)
PYTHONPATH=src .venv/bin/python scripts/provision_grafana.py

# 3. a clearance from the terminal
PYTHONPATH=src .venv/bin/python scripts/run_pipeline.py docs/samples/test_ad.mp4 FR SA US

# 4. or run Launch Control
PYTHONPATH=src .venv/bin/uvicorn customs.app:app --reload
# open http://127.0.0.1:8000

# 5. tests: offline by default, -m live hits Gemini and Grafana
.venv/bin/python -m pytest -q
```

Cloud Run

```
scripts/deploy.sh          # idempotent: APIs, secrets, IAM, deploy, webhook wiring
```

The deploy script enables the required GCP APIs, stores the two Grafana tokens in Secret Manager (piped on stdin, never in shell history), grants the runtime service account only `aiplatform.user` plus accessor on those two secrets, deploys a single instance (SQLite and in-process locks are a stated demo-grade tradeoff), and repoints the Grafana webhook contact point at the new service URL. It prints the service URL as its last line.

Honest limits, and where everything is

Limits, stated on purpose

- ◆ **The Guard is deliberate friction.** Protected-basis and offence-class findings are never auto-remediated; they are cited and routed to a human.
- ◆ **Unsourced findings are capped.** No live citation means severity capped at 40, no remediation, no market block. Hallucination is rendered as a visible quality signal, not a hidden risk.
- ◆ **Input caps.** 120 seconds and 200 MB per upload, enforced at the door.
- ◆ **Single instance by design.** SQLite, in-process locks, Cloud Run `--max-instances 1` . Demo-grade persistence, stated tradeoff.
- ◆ **15 markets, not 195.** The pack format is the extension point, and the packs are data.

Resources

RESOURCE	WHERE
Repository	github.com/tdries/customs
Live instance	customs-app-akap4ao72a-ew.a.run.app
Design document	docs/superpowers/specs/2026-08-23-customs-design.md
Milestone gate report	docs/superpowers/plans/milestone1-gate.md
Landmine ground truth	docs/samples/landmines.yaml
Devpost copy	docs/devpost.md
Demo video script	docs/demo-video-script.md
License	Apache-2.0, LICENSE

Repository layout

src/customs/	agents, console, telemetry, store	markets/	15 packs + taxonomy
grafana/	dashboard definitions	scripts/	pipeline, provision, deploy
tests/	339 offline + live suite	docs/	spec, plan, screenshots, manual



**A statute behind every finding.
A master that can actually ship.**

github.com/tdries/customs

customs-app-akap4ao72a-ew.a.run.app

Agentic Cinema 2026 / Grafana Labs track / Apache-2.0